

BANK DATABASE – ACID PROPERTIES & ISOLATION

2520030030

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1. Create Database

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window contains the following SQL script:

```
1 • CREATE DATABASE Bank_ACID_DB1;  
2 • USE Bank_ACID_DB1;
```

The 'Output' window shows the execution results:

#	Time	Action	Message
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected

2. Create Customer Table

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window contains the following SQL script:

```
1 • CREATE DATABASE Bank_ACID_DB1;  
2 • USE Bank_ACID_DB1;  
3 • CREATE TABLE Customer (  
4     Customer_ID INT PRIMARY KEY,  
5     Customer_Name VARCHAR(100) NOT NULL,  
6     Phone VARCHAR(15),  
7     City VARCHAR(50)  
8 );
```

The 'Output' window shows the execution results:

#	Time	Action	Message
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
✓ 3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL....	0 row(s) affected

3. Create Account Table

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window contains the following SQL script:

```
9 • CREATE TABLE Account (  
10     Account_No INT PRIMARY KEY,  
11     Customer_ID INT,  
12     Account_Type VARCHAR(20),  
13     Balance DECIMAL(12,2),  
14     Branch VARCHAR(50),  
15     FOREIGN KEY (Customer_ID)  
16     REFERENCES Customer(Customer_ID)  
17 );
```

The 'Output' window shows the execution results:

#	Time	Action	Message
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
✓ 3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL....	0 row(s) affected
✓ 4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected

4. Create Bank Transaction Table

```
18 • CREATE TABLE Bank_Transaction (  
19     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,  
20     Account_No INT,  
21     Transaction_Type VARCHAR(20),  
22     Amount DECIMAL(12,2),  
23     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,  
24     FOREIGN KEY (Account_No)  
25     REFERENCES Account(Account_No)  
26 );
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected	
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected	
✓ 3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL...	0 row(s) affected	
✓ 4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected	
✓ 5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected	

5. Insert Customer Data

```
27 • INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES  
28 (101, 'Rohit Sai', '9876543210', 'Hyderabad'),  
29 (102, 'Virat', '9876543211', 'Benguluru'),  
30 (103, 'Rohit Sharma', '9876543212', 'Mumbai'),  
31 (104, 'NTR', '9876543213', 'Hyderabad'),  
32 (105, 'Prabhas', '9876543214', 'Hyderabad');
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected	
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected	
✓ 3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL...	0 row(s) affected	
✓ 4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected	
✓ 5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected	
✓ 6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'Hy...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	

6. Insert Account Data

```
33 • INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES  
34 (10001, 101, 'Savings', 50000, 'Hyderabad'),  
35 (10002, 102, 'Savings', 75000, 'Benguluru'),  
36 (10003, 103, 'Current', 120000, 'Mumbai'),  
37 (10004, 104, 'Savings', 45000, 'Hyderabad'),  
38 (10005, 105, 'Current', 90000, 'Hyderabad');
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected	
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected	
✓ 3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL...	0 row(s) affected	
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✓ 6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'Hy...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	
✓ 7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savi...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	

7. Insert Bank Transaction Data

```
39 • INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES
40 (10001, 'DEPOSIT', 10000),
41 (10002, 'DEPOSIT', 15000),
42 (10003, 'WITHDRAW', 20000),
43 (10004, 'DEPOSIT', 5000),
44 (10005, 'WITHDRAW', 10000);
```

Output

Action Output

#	Time	Action	Message
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
✓ 3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL,...	0 row(s) affected
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✓ 6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'Hy...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savi...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 8	18:51:32	INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

8. Display Data

Query 1 physician Joins SQL File 5* Triggers & Procedures* Acid* x

Limit to 1000 rows

```
46 • SELECT * FROM Customer;
47
48
49
50
```

Result Grid

Customer_ID	Customer_Name	Phone	City
101	Rohit Sai	9876543210	Hyderabad
102	Virat	9876543211	Benguluru
103	Rohit Sharma	9876543212	Mumbai
104	NTR	9876543213	Hyderabad
105	Prabhas	9876543214	Hyderabad
* NULL	NULL	NULL	NULL

Output

Action Output

#	Time	Action	Message
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
✓ 3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
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✓ 7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sa...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 8	18:51:32	INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 9	18:52:23	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned

48 • `SELECT * FROM Account;`

49

50

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Benguluru
10003	103	Current	120000.00	Mumbai
10004	104	Savings	45000.00	Hyderabad
10005	105	Current	90000.00	Hyderabad

Account 2 x

Output

Action Output

#	Time	Action	Message
1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
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5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected
6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'H...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sa...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	18:51:32	INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	18:52:23	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	18:53:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned

50 • `SELECT * FROM Bank_Transaction;`

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-26 18:51:32
2	10002	DEPOSIT	15000.00	2026-08-26 18:51:32
3	10003	WITHDRAW	20000.00	2026-08-26 18:51:32
4	10004	DEPOSIT	5000.00	2026-08-26 18:51:32
5	10005	WITHDRAW	10000.00	2026-08-26 18:51:32

Bank_Transaction 3 x

Output

Action Output

#	Time	Action	Message
1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCH...	0 row(s) affected
5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected
6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'H...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sa...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
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9	18:52:23	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	18:53:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	18:54:10	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned

9. Start Transaction

52 • `START TRANSACTION;`

53

Output

Action Output

#	Time	Action	Message
1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCH...	0 row(s) affected
5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected
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7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sa...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	18:51:32	INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	18:52:23	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	18:53:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	18:54:10	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	18:54:53	START TRANSACTION	0 row(s) affected

10. Deposit Money

```
54 • UPDATE Account
55 SET Balance = Balance + 5000
56 WHERE Account_No = 10001;
57
```

Output

Action Output

#	Time	Action	Message
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
✓ 3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
✓ 4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected
✓ 5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected
✓ 6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'H...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sa...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 8	18:51:32	INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 9	18:52:23	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓ 10	18:53:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 11	18:54:10	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 12	18:54:53	START TRANSACTION	0 row(s) affected
✓ 13	18:56:26	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

11. Check Balance

```
57
58 • SELECT *
59 FROM Account
60 WHERE Account_No = 10001;
61
```

Limit to 1000 rows

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

12. Commit Transaction

```
62 • COMMIT;
63
```

Output

Action Output

#	Time	Action	Message
✓ 1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
✓ 2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
✓ 3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
✓ 4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected
✓ 5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected
✓ 6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'H...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sa...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 8	18:51:32	INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 9	18:52:23	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓ 10	18:53:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 11	18:54:10	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 12	18:54:53	START TRANSACTION	0 row(s) affected
✓ 13	18:56:26	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 14	18:57:27	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
✓ 15	18:58:05	COMMIT	0 row(s) affected

13. Check Balance After Commit

```
63 • SELECT *
64 FROM Account
65 WHERE Account_No = 10001;
66
67
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

14. Start Transaction for Rollback

```
67 • START TRANSACTION;
68
```

#	Time	Action	Message
1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected
5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected
6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'H...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sa...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	18:51:32	INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	18:52:23	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	18:53:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	18:54:10	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	18:54:53	START TRANSACTION	0 row(s) affected
13	18:56:26	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	18:57:27	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
15	18:58:05	COMMIT	0 row(s) affected
16	18:58:55	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
17	18:59:42	START TRANSACTION	0 row(s) affected

15. Withdraw Money

```
68 • UPDATE Account
69 SET Balance = Balance - 10000
70 WHERE Account_No = 10001;
```

#	Time	Action	Message
1	18:43:36	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected
2	18:43:38	USE Bank_ACID_DB1	0 row(s) affected
3	18:44:27	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected
5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected
6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'H...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sa...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	18:51:32	INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	18:52:23	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	18:53:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	18:54:10	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	18:54:53	START TRANSACTION	0 row(s) affected
13	18:56:26	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	18:57:27	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
15	18:58:05	COMMIT	0 row(s) affected
16	18:58:55	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
17	18:59:42	START TRANSACTION	0 row(s) affected
18	19:00:26	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

16. Check Balance

```
71 • SELECT *
72 FROM Account
73 WHERE Account_No = 10001;
74
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

17. Rollback and Check Balance

```
74 • ROLLBACK;
75 • SELECT *
76 FROM Account
77 WHERE Account_No = 10001;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

18. Start Transaction for Savepoint

Query 1 physician Joins SQL File 5*

78 • START TRANSACTION;

22 19:03:37 START TRANSACTION 0 row(s) affected

19. First Update

```
80 • UPDATE Account
81 SET Balance = Balance + 5000
82 WHERE Account_No = 10001;
83
```

Output

Action Output

#	Time	Action	Message
4	18:45:14	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(50))	0 row(s) affected
5	18:46:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Amount DECIMAL(10,2))	0 row(s) affected
6	18:49:00	INSERT INTO Customer(Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Rohit Sai', '9876543210', 'Hyderabad')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	18:50:29	INSERT INTO Account(Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 45000.00, 'Hyderabad')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	18:51:32	INSERT INTO Bank_Transaction(Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000.00)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	18:52:23	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	18:53:25	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	18:54:10	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	18:54:53	START TRANSACTION	0 row(s) affected
13	18:56:26	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	18:57:27	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
15	18:58:05	COMMIT	0 row(s) affected
16	18:58:55	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
17	18:59:42	START TRANSACTION	0 row(s) affected
18	19:00:26	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
19	19:01:52	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
20	19:02:48	ROLLBACK	0 row(s) affected
21	19:02:50	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
22	19:03:37	START TRANSACTION	0 row(s) affected
23	19:05:33	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

20. Create First Savepoint

Query 1	physician	Joins	SQL File 5*	Triggers & Procedures*	Acid*	
						
84 • SAVEPOINT Deposit1;						
✓	24	19:06:54	SAVEPOINT Deposit1	0 row(s) affected		0.000 sec

21. Second Update

Query 1	physician	Joins	SQL File 5*	Triggers & Procedures*	Acid*	
						
84 • SAVEPOINT Deposit1;						
85 • UPDATE Account						
86 SET Balance = Balance - 3000						
87 WHERE Account No = 10002;						
✓	24	19:06:54	SAVEPOINT Deposit1	0 row(s) affected		0.000 sec
✓	25	19:08:29	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0		0.000 sec

22. Create Second Savepoint

Query 1	physician	Joins	SQL File 5*	Triggers & Procedures*	Acid*	
						
84 • SAVEPOINT Deposit1;						
85 • UPDATE Account						
86 SET Balance = Balance - 3000						
87 WHERE Account_No = 10002;						
88 • SAVEPOINT Withdrawal1;						
✓	24	19:06:54	SAVEPOINT Deposit1	0 row(s) affected		0.000 sec
✓	25	19:08:29	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0		0.000 sec
✓	26	19:09:45	SAVEPOINT Withdrawal1	0 row(s) affected		0.000 sec

23. Rollback to Second Savepoint

89						
90 • ROLLBACK TO SAVEPOINT Withdrawal1;						
✓	25	19:08:29	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0		0.000 sec
✓	26	19:09:45	SAVEPOINT Withdrawal1	0 row(s) affected		0.000 sec
✓	27	19:11:09	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected		0.000 sec

24. Commit

90 • ROLLBACK TO SAVEPOINT Withdrawal1;						
91 • COMMIT;						
92						
93						
✓	25	19:08:29	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0		0.000 sec
✓	26	19:09:45	SAVEPOINT Withdrawal1	0 row(s) affected		0.000 sec
✓	27	19:11:09	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected		0.000 sec
✓	28	19:12:06	COMMIT	0 row(s) affected		0.000 sec

25. Check Accounts

92 • `SELECT * FROM Account;`

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: [IA](#)

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	60000.00	Hyderabad
	10002	102	Savings	72000.00	Benguluru
	10003	103	Current	120000.00	Mumbai
	10004	104	Savings	45000.00	Hyderabad
	10005	105	Current	90000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL